

HIGHLIGHTS

- Black or African American/Other patients may be more likely to miss scheduled appointments than White patients.
- Black or African American/Other patients were exposed to certain harmful air particles at a higher rate than White patients.
- Few patients experienced an exacerbation or healthcare utilization in the first year of biologic therapy. No differences in response outcomes were seen by race.

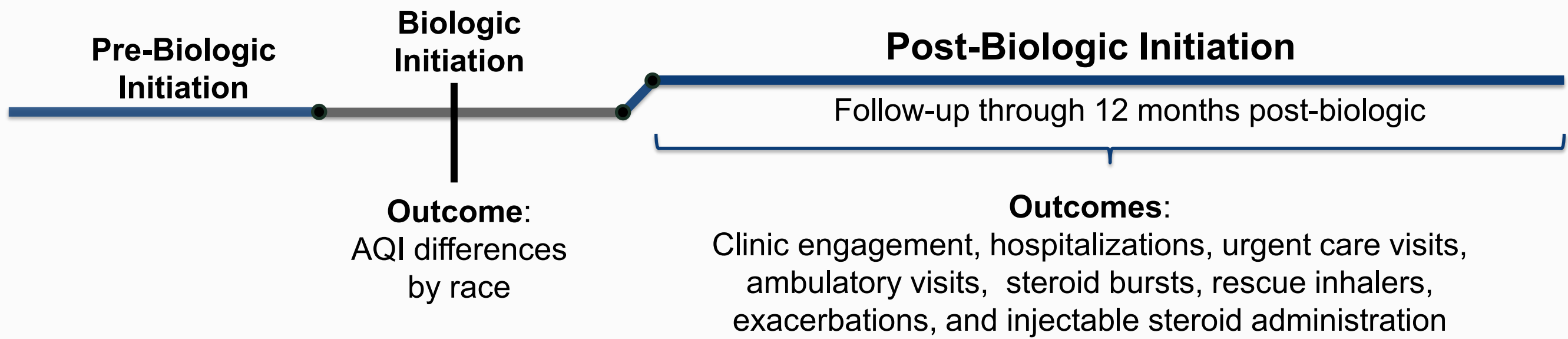
PURPOSE

This study described clinical outcomes in patients with moderate to severe asthma in the first year of biologic therapy and evaluated air quality and clinic engagement by race.

METHODS

- Design
 - Retrospective cohort study
 - Enrollment: 1/1/2019-6/20/2022 with a 12-month follow up through 6/30/2023
 - Patients were geocoded then matched with air quality index (AQI) data from the Environmental Protection Agency
- Sample
 - Inclusion criteria: initiated a new biologic (benralizumab, dupilumab, mepolizumab, and omalizumab) between 1/1/2019 and 6/30/2022
 - Exclusion criteria:
 - Indication other than moderate to severe asthma
 - Biologic not prescribed by a Vanderbilt Allergy/ Immunology Clinic provider

Figure 1. Study Design



RESULTS

Baseline Demographics (N=185)

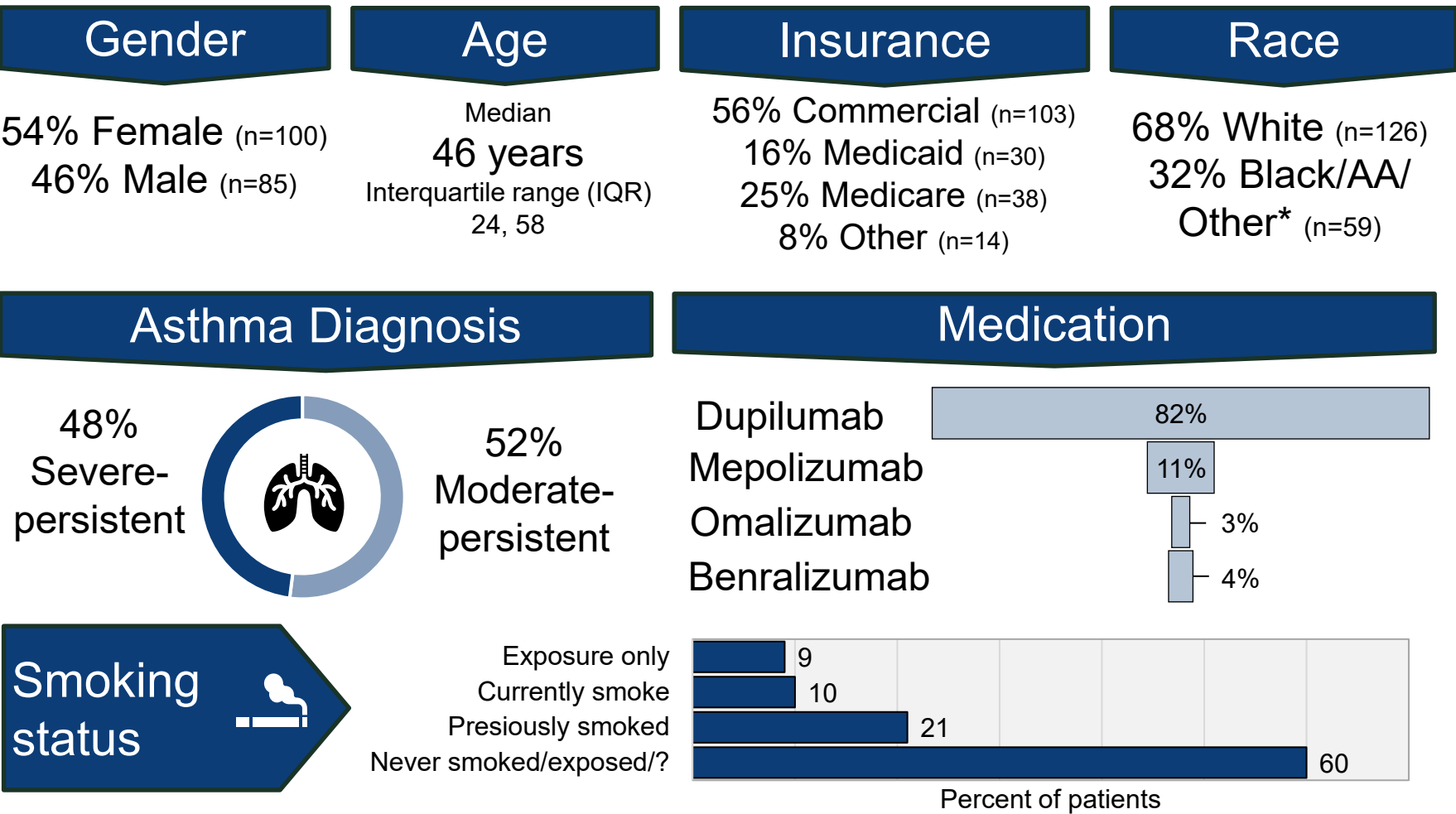
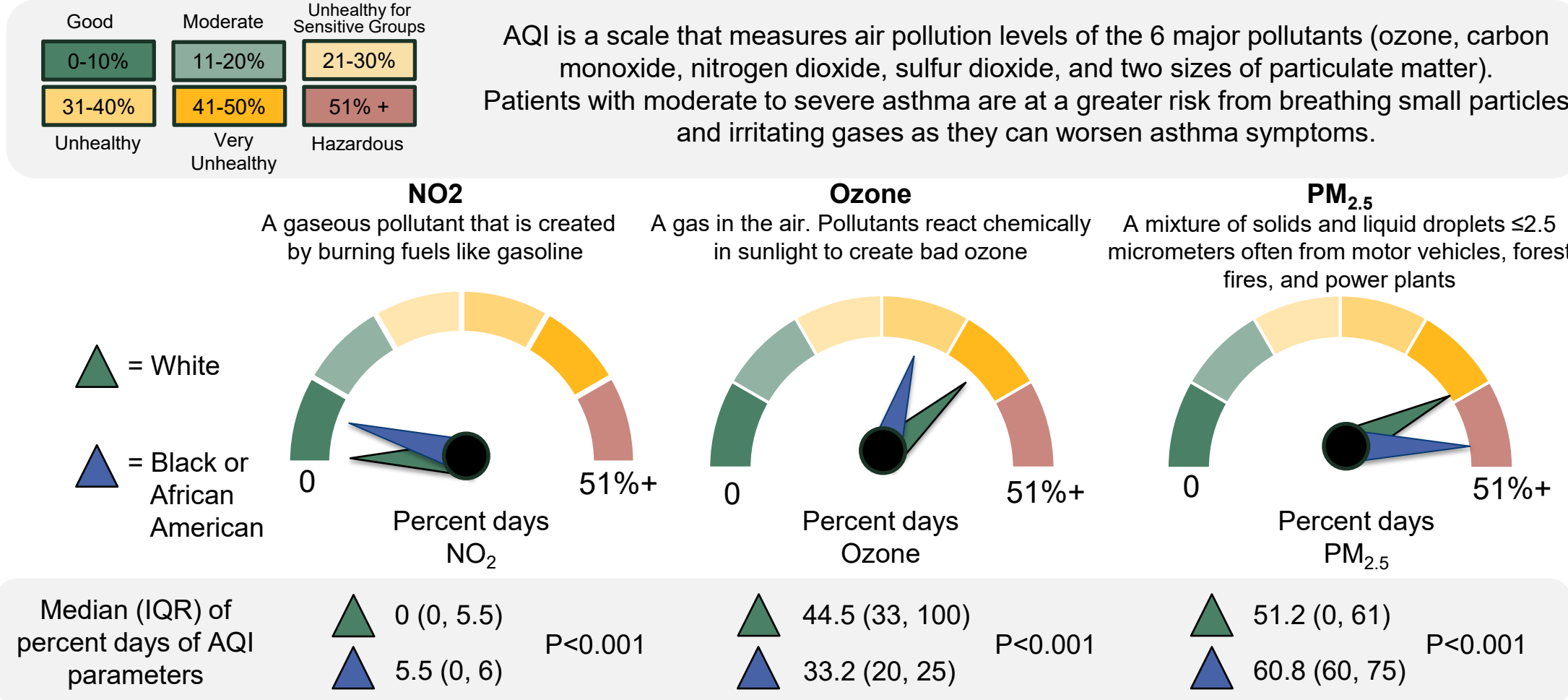


Figure 2. Air Quality Index Differences by Race



Clinic Engagement

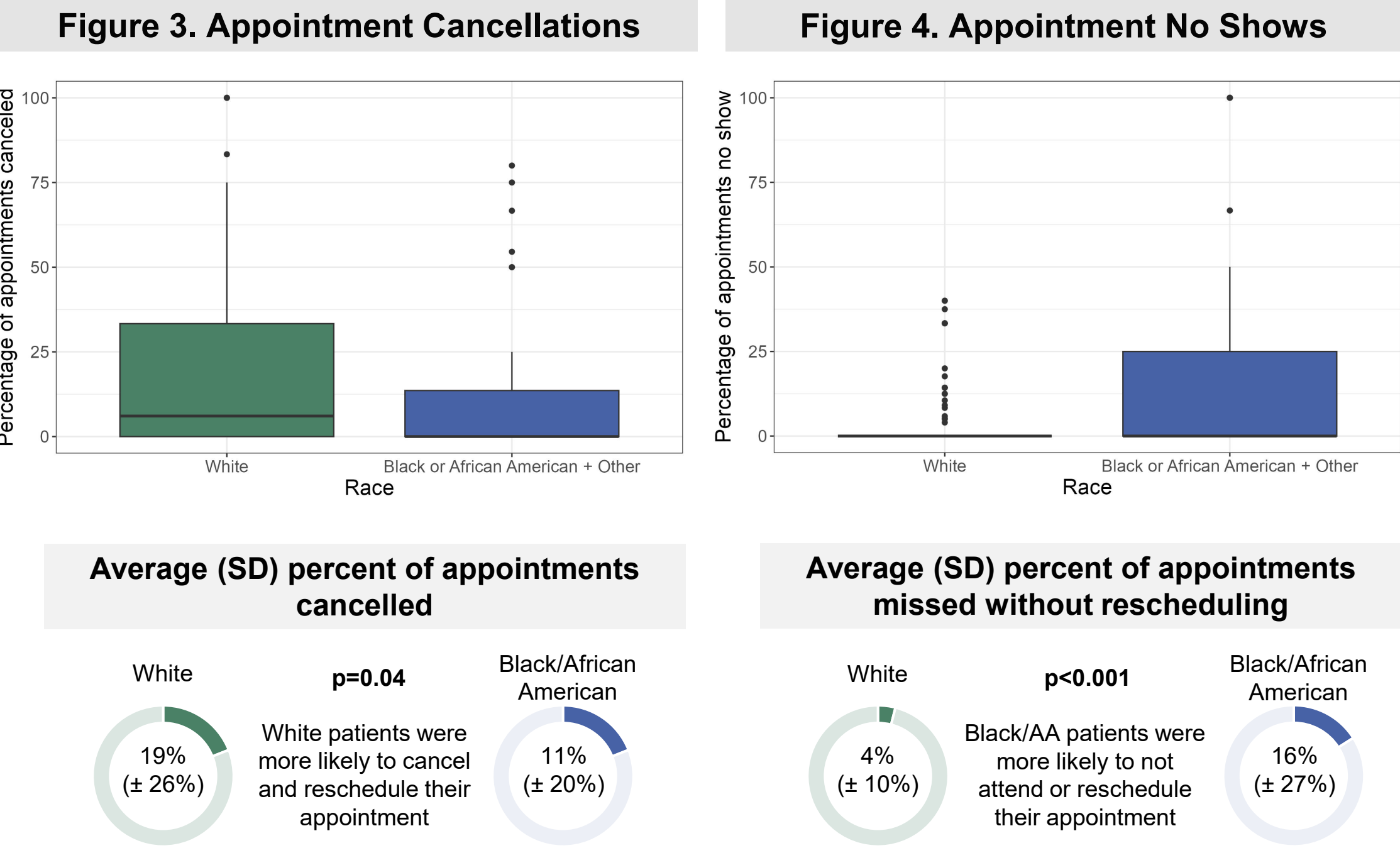


Table 1. Clinical Outcomes 12 months from Biologic Initiation

	Frequency of Outcomes n (%)			
	0	1	2	3+
Hospitalizations	171 (92)	10 (5)	3 (2)	1 (<1)
Urgent care visits	168 (91)	14 (8)	2 (1)	1 (<1)
Ambulatory visits	150 (81)	25 (14)	9 (5)	1 (<1)
Steroid bursts	149 (81)	26 (14)	6 (3)	4 (2)
Rescue inhalers	106 (57)	15 (8)	9 (5)	55 (30)
Injectable steroid administrations	181 (98)	4 (2)	0 (0)	0 (0)
Exacerbations	120 (65)	29 (16)	18 (10)	18 (10)

Differences in clinical outcomes were not significantly different by race.

References: Air Pollutants. Centers for Disease Control and Prevention. Accessed October 22, 2024. <https://www.cdc.gov/air-quality/pollutants/index.html>.

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Conflicts of Interest: None related to this work.

*Other race: American Indian/Alaskan Native, Native Hawaiian/other Pacific Islander, Middle Eastern, Mixed race